into a tube, salver or funnel shaped; twisted or rarely valvate, often hairy within or contain some corona like out growths at the mouth of the corolla tube.

Androecium: Stamens 5 , alternipetalous, often epipetalous, apostemonous to monadelphous, In *Asclepiadoids* the stamens are connate to the styles to form a **gynostegium**, pollen grains of each theca of an anther are fused into a waxy mass called **pollinium**. The right pollinium of each anther attached to the left pollinium of the adjacent anther by a hair like translator, translator arms (**retinacula**) attached together with the gland like structure called **corpusculum**. Anthers are dithecous, basifixed, often sagitate, introse; dehisce longitudinally, anthers basally awned; sometimes bear hairy appendages over the lobes (*Nerium*).

Gynoecium: Bicarpellary, carpels apically united, superior, or rarely half inferior (*Plumeria*) 1 to 2 locule with 2 to many ovules in each locule on marginal placentation. Style one and simple, stigma is characteristically thickened, massive and bilobed. A nectariferous disc is often present around or at the base of the gynoecium, (*Thevetia, Catharanthus, Allamanda* and *Rauvolfia*).



Fruit: The fruit is variable and can be a berry (*Landolphia*), drupe (*Cerbera*) follicle (*Asclepias*), capsule (*Allamanda*).

Seed: Seeds are endospermous often with crown of hairs.

Botanical description of *Catharanthus roseus*

Habit: Erect ever blooming ornamental plant with milky latex.

Root: Branched tap root system

Stem: Aerial, erect, cylindrical reddish green, glabrous and branched.

Leaves: Usually simple, opposite decussate, exstipulate, subsessile, or petiolate, elliptic – ovate, entire, mucronate, unicostate reticulate venation.

Inflorescence: cymose, axillary pairs.

Flower: Ebracteate, Ebracteolate, subsessile, complete, bisexual, heterochlamydeous, actinomorphic, hypogynous, pentamerous, rosy purple, white or pink.

Calyx: Sepals 5, slightly synsepalous, green showing valvate aestivation.

Corolla: Petals 5, sympetalous, throat of corolla tube hairy forming a corona, twisted (hypocrateriform).

Androecium: Stamens 5, apostemanous, epipetalous, inserted at the mouth of the corolla tube, filaments short, anthers sagittate, dithecous, dorsifixed, introse.

Gynoecium: Bicarpellary, apocarpous, ovaries superior, unilocular, ovules many, placentation marginal, style simple, stigma hour-glass shaped. Two scaly nectaries are present one on the anterior and another on the posterior side of the ovary.

Fruit: A pair of elongated follicles.

Floral Formula: $\text{Ebr.,Ebrl.,}\oplus,\ \male\female,K_{(5)},\overset{\frown}{C_{(5)}},A_5,\underline{G}_{(2)}$